

Abstract

This project presents the design and implementation of a smart clap-controlled lighting system using linear integrated circuits. The system detects sound signals through an electret condenser microphone, amplifies them using the LM358N operational amplifier, and converts the amplified signal into a digital pulse using a comparator stage also implemented with the LM358N. A NE555P timer configured in monostable mode generates a clean, debounced pulse which is fed as a clock signal to a CD4013 D flip-flop wired in toggle configuration. Each successive clap alternately turns the LED indicator ON and OFF, demonstrating practical applications of op-amps, comparators, timers, and digital sequential logic circuits.

The amplifier stage uses a non-inverting configuration with a gain of approximately 471 ($R_f = 470\Omega$, $R_{in} = 1\text{ k}\Omega$), sufficient to raise the microphone's millivolt-level output to a level that reliably drives the comparator. The comparator compares the amplified audio signal against a reference voltage set by a 100K and 10K voltage divider, producing a low-going output pulse when a clap is detected. The NE555 monostable stage, with $R = 47\text{ k}\Omega$ and $C = 10\text{ }\mu\text{F}$, generates a clean output pulse of approximately 0.52 seconds, suppressing double-triggering from clap echoes. The CD4013 flip-flop toggles its Q output with each clock edge, driving the LED through a current-limiting resistor.

The circuit operates on a single 9 V DC battery supply and demonstrates fundamental concepts including operational amplifier gain, voltage comparison, monostable pulse generation, and D flip-flop toggle operation. The system was tested and verified to reliably toggle the LED on consecutive claps with no observable false triggering.

Keywords: LM358N, Operational Amplifier, NE555 Timer, CD4013 Flip-Flop, Monostable, Comparator, Clap Detection, Sound-Activated Switch, Toggle Circuit.

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1 Introduction

Sound-activated switching systems represent a practical and intuitive approach to hands-free control of electrical appliances. By detecting audio events such as claps, these systems enable users to control lighting and other loads without physical interaction with a switch. This is particularly useful in accessibility applications, smart home environments, and energy management scenarios where reducing unnecessary manual operation is desirable.

The present project implements a clap-controlled LED toggle switch using a cascade of well-established analog and digital integrated circuits. An electret condenser microphone captures the acoustic signal produced by a clap, converts it into a small alternating voltage, and this signal is then processed through amplification, comparison, pulse shaping, and digital toggling stages, ultimately controlling the state of a light-emitting diode.

The LM358N dual op-amp IC provides both the amplification and comparison functions. The NE555P timer, configured in monostable mode, eliminates multiple triggering caused by clap echoes and mechanical noise. The CD4013 D flip-flop performs the toggle function, maintaining the ON or OFF state of the output between clap events. The entire system operates from a single 9 V battery, making it compact, portable, and suitable for battery-powered applications.

This project serves as a comprehensive demonstration of fundamental concepts covered in the Linear Integrated Circuits course: non-inverting amplifier design, voltage comparator operation, 555 timer monostable mode, and D flip-flop toggle configuration, all integrated into a functioning real-world system.

2 Objectives

1. To design a clap-activated sound detection circuit using an electret condenser microphone.
2. To amplify weak microphone output signals using the LM358N op-amp in non-inverting configuration.
3. To convert the amplified analog signal to a digital pulse using an inverting comparator stage.
4. To set the detection sensitivity using a fixed reference voltage generated by a 100 k Ω and 10 k Ω voltage divider.
5. To generate a clean, debounced monostable pulse using the NE555P timer.
6. To implement a toggle switching mechanism using the CD4013 D flip-flop wired in toggle mode.
7. To drive a 5 mm LED as a visual indicator of the system's output state.
8. To operate the entire circuit from a single 9 V DC battery supply.
9. To experimentally verify the circuit behaviour and confirm reliable ON/OFF toggling with consecutive claps.

10. To develop practical skills in breadboard circuit assembly, component selection, and systematic testing.

3 Components Required

Table 1 lists all components used in this project along with their specifications and quantities.

Table 1: Bill of Materials

Component	Specification / Value	Qty
Operational Amplifier	LM358N, Dual Op-Amp, DIP-8	1
Timer IC	NE555P, DIP-8	1
D Flip-Flop IC	CD4013BE, Dual D Flip-Flop, DIP-14	1
Microphone	Electret Condenser, 2-pin, 9–12 mm	1
LED	5 mm, standard colour	1
Potentiometer	B470K	1
Resistor R1 (Mic bias)	10 k Ω , 1/4 W, 5%	1
Resistor R2 (Amp input)	1 k Ω , 1/4 W, 5%	1
Resistor Rf (Amp feedback)	470 Ω , 1/4 W, 5%	1
Resistor R4 (Comp ref)	100 k Ω , 10k 1/4 W, 5%	1
Resistor R5 (LED current)	1 k Ω , 1/4 W, 5%	1
Resistor R6 (555 timing)	47 k Ω , 1/4 W, 5%	1
Capacitor C1 (mic bypass)	0.01 μ F, 50 V ceramic	1
Capacitor C2 (filter)	0.1 μ F, 50 V ceramic	1
Capacitor C3 (555 timing)	10 μ F, 25 V electrolytic	1
Capacitor C4 (555 bypass)	0.01 μ F, 50 V ceramic	1
Battery	9 V PP3/6F22, alkaline	1
Battery Connector	9 V snap connector	1
Breadboard	830-tie solderless breadboard	1
Jumper Wires	Solid-core, 22 AWG	As reqd.
Multimeter	For voltage verification	1

4 Theoretical Background

4.1 The LM358N Operational Amplifier

The LM358N is a dual general-purpose operational amplifier designed for single-supply operation from 3 V to 32 V. It contains two independent, high-gain, internally compensated op-amp stages. Unlike the older LM741, the LM358 is specifically designed to operate with its inputs and outputs close to ground, making it ideal for single-supply applications. Key characteristics include an open-

loop gain of approximately 100 dB, a unity-gain bandwidth of 1 MHz, and a common-mode input voltage range that includes ground.

In this project, the LM358N serves two distinct roles: one half is used as a non-inverting amplifier, and the other half is used as an inverting comparator. The dual-op-amp package reduces component count and breadboard space.

Table 2: Selected LM358N Electrical Characteristics

Parameter	Symbol	Typical Value
Supply voltage range	VS	3 V – 32 V (single supply)
Open-loop voltage gain	AOL	100 dB (100,000 V/V)
Unity-gain bandwidth	GBW	1 MHz
Input offset voltage	VOS	2 mV
Input bias current	IB	45 nA
Common-mode input range	VICR	0 V to VS – 1.5 V
Output voltage swing	VO	0 V to VS – 1.5 V
Slew rate	SR	0.6 V/ μ s
Quiescent current (per amp)	IQ	0.5 mA

4.2 Non-Inverting Amplifier Stage

The first op-amp (LM358N, section A) is configured as a non-inverting amplifier. The electret microphone signal is applied to the non-inverting input (pin 3) through a 0.1 μ F coupling capacitor, which blocks any DC bias from the microphone and passes only the AC audio signal. A 10 k Ω resistor provides the microphone's DC bias current path to the supply.

The voltage gain of the non-inverting amplifier is given by:

$$A_v = 1 + (R_f / R_1) = 1 + (470\text{k}\Omega / 1\text{ k}\Omega) = 471$$

This gain of 471 amplifies the microphone's typical output of a few millivolts to several hundred millivolts, sufficient to reliably drive the comparator stage. The 1 k Ω input resistor (R2) also serves to limit the input bias current path and stabilise the DC operating point.

4.3 Inverting Comparator Stage

The second op-amp (LM358N, section B) is configured as an inverting comparator operating without negative feedback. The amplified signal from the first stage is applied to the inverting input (pin 6). A reference voltage is generated using a voltage divider consisting of a 100 k Ω resistor connected to Vcc and a 10 k Ω resistor connected to ground. The junction of these resistors is applied to the non-inverting input (pin 5) of the op-amp comparator. When the clap signal at the

inverting input exceeds the reference voltage, the comparator output (pin 7) switches to a low state (near 0 V). When the signal falls below the reference, the output returns to a high state (near $V_S - 1.5$ V). This low-going pulse is compatible with the active-low trigger input of the NE555 timer.

- The inverting comparator is preferred in this design because it produces a **LOW output pulse**, which is directly compatible with the triggering requirement of the 555 timer.

4.4 NE555 Timer in Monostable Mode

The NE555P timer is configured in monostable mode, acting as a one-shot pulse generator. When a negative-going pulse is applied to its trigger input (pin 2), the output (pin 3) goes HIGH for a fixed duration determined by the external RC network ($R = 47\text{k}\Omega$, $C = 10\text{ }\mu\text{F}$) and then returns LOW automatically. The output pulse width is given by:

$$T = 1.1 \times R \times C = 1.1 \times 47,000 \times 10 \times 10^{-6} \approx 0.52 \text{ s}$$

This approximately 0.52-second pulse ensures that the flip-flop receives a single clean rising edge per clap, regardless of echo-induced multiple pulses from the comparator. The $0.01\text{ }\mu\text{F}$ bypass capacitor at the control voltage pin (pin 5) filters supply noise and improves timing stability. Pin 4 (RESET) is tied to VCC to prevent accidental resets. A $0.1\text{ }\mu\text{F}$ filter capacitor at the comparator output also attenuates noise before it reaches the 555 trigger.

4.5 CD4013 D Flip-Flop in Toggle Mode

The CD4013 is a CMOS dual D flip-flop. In this project, one flip-flop section is configured in toggle mode by connecting the complement output (Q-bar) back to the Data input (D). As a result, each rising edge on the clock input (CLK) causes the Q output to change state: if Q was HIGH, it becomes LOW, and vice versa. This behaviour directly implements the toggle function required for the clap-to-toggle control.

The NE555 timer output provides the clock signal to the CLK pin. The Set and Reset pins of the CD4013 are grounded to disable asynchronous overrides. The Q output drives the LED through a $1\text{ k}\Omega$ current-limiting resistor, ensuring an LED forward current of approximately 5–8 mA, well within safe limits.

5 Methodology

5.1 Design Phase

The design began with a functional block decomposition: sensing, amplification, comparison, pulse shaping, toggling, and output indication. Component values were determined analytically. The

amplifier gain of 471 was chosen to reliably raise microphone signals from the millivolt range to several hundred millivolts. The 10 k Ω bias resistor for the microphone was selected to ensure proper electret capsule biasing without excessive current draw.

The comparator reference voltage is connected to 100k and 10k resistor, allowing the sensitivity of the circuit to be tuned to the ambient noise environment. The NE555 timing components (47k Ω , 10 μ F) were chosen to produce a pulse of approximately 0.52 seconds — long enough to constitute a definitive clock edge for the flip-flop, and short enough to allow two rapid claps to be distinguished as separate toggle events.

5.2 Circuit Assembly

11. The LM358N IC was inserted straddling the DIP groove of the breadboard. Pin functions were verified against the DIP-8 datasheet pinout.
12. The microphone was wired with its positive terminal (pin with ground ring) connected via a 10 k Ω bias resistor to the 9 V supply rail. The output was coupled to pin 3 of the LM358N amplifier section through a 0.1 μ F capacitor.
13. Feedback resistor R_f (470k Ω) was connected between the amplifier output (pin 1) and the inverting input (pin 2). Input resistor R₁ (1 k Ω) was placed between pin 2 and the microphone coupling capacitor.
14. The comparator section (pins 5, 6, 7 of LM358N) was wired with the amplifier output connected to pin 6 (inverting input). The 100K and the 10k(voltage divider) was connected to pin 5 (non-inverting input) to set the adjustable reference.
15. The NE555P was inserted and wired: pin 8 to VCC, pin 1 to GND, pin 4 (RESET) to VCC, pin 2 (TRIG) to the comparator output via a 0.1 μ F filter capacitor, pin 5 (CV) to GND via a 0.01 μ F bypass capacitor, and the RC network (47k Ω and 10 μ F) between pins 6/7 and supply/ground.
16. The CD4013 flip-flop was inserted with pin 3 (CLK) connected to the NE555 output (pin 3). Q-bar (pin 2) was connected back to D (pin 5). Set and Reset (pins 6, 4) were grounded. Q (pin 1) was connected through a 1 k Ω resistor and then to the LED anode; the LED cathode was grounded.
17. Power connections were double-checked before connecting the battery.

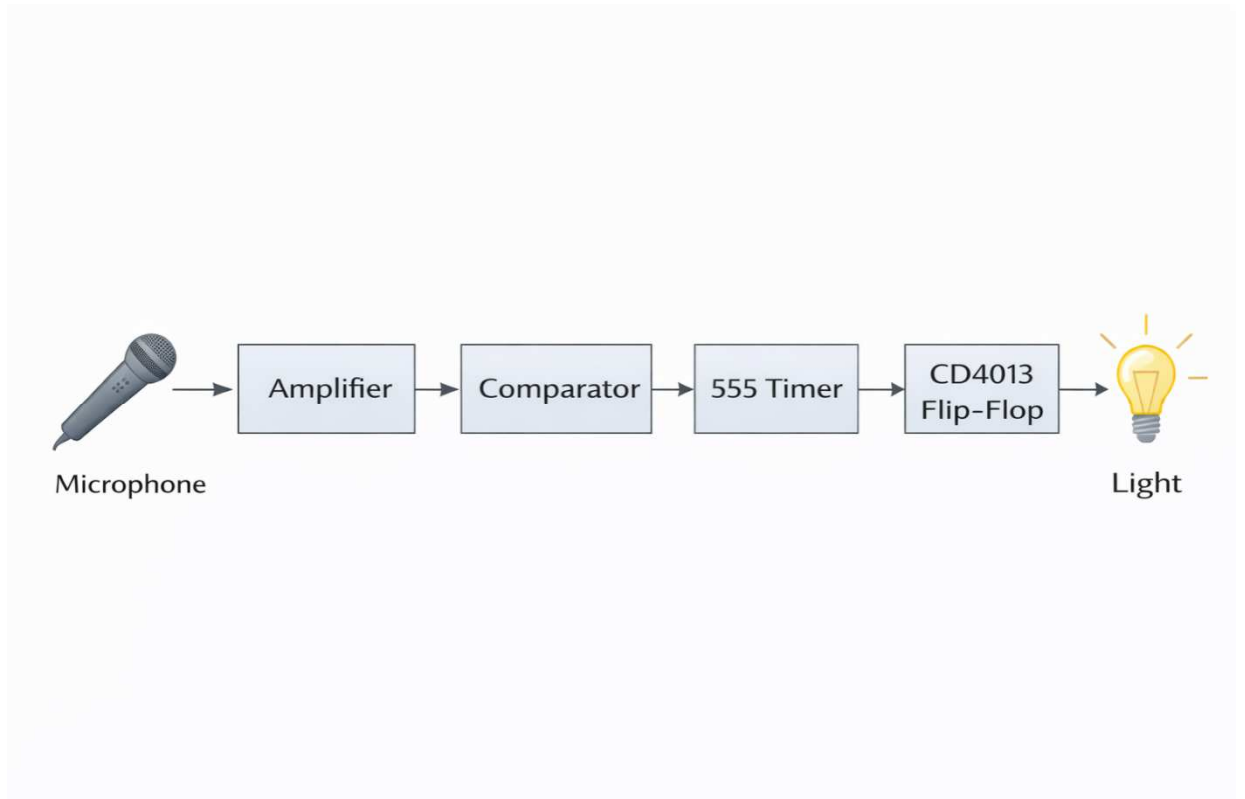
5.3 Testing and Verification

18. The amplifier output was monitored with a multimeter to confirm amplification of microphone signals when a clap was produced nearby.
19. The comparator output was verified to produce a low-going pulse upon each clap detection. The potentiometer was adjusted to achieve reliable triggering without false activation from ambient noise.
20. The NE555 output was confirmed to produce a single, clean HIGH pulse of approximately 1 second following each clap trigger.
21. The LED was observed to toggle ON and OFF with each successive clap. Ten consecutive toggle cycles were performed without false triggering or missed transitions.

6 Block Diagram

The system comprises five cascaded functional blocks:

- Sensing Block: The electret condenser microphone with its bias resistor and coupling capacitor. Converts acoustic clap energy into a small alternating electrical signal.
- Amplifier Block: LM358N (section A) in non-inverting configuration (gain = 471). Raises the microphone signal to a level suitable for comparison.
- Comparator Block: LM358N (section B) in open-loop inverting comparator mode generates a low-going digital pulse upon clap detection.
- Pulse Shaping Block: NE555P in monostable mode. Produces a clean, fixed-width HIGH pulse from the comparator's trigger output, eliminating echo-induced double-triggering.
- Toggle & Output Block: CD4013 D flip-flop wired in toggle mode. Alternates the Q output between HIGH and LOW on each clock pulse, driving the LED indicator through current-limiting resistors.



7 Circuit Diagram

Figure 1 shows the complete circuit schematic of the clap-controlled room light system. The signal flow proceeds from left to right: microphone → amplifier → comparator → NE555 monostable → CD4013 toggle → LED output.

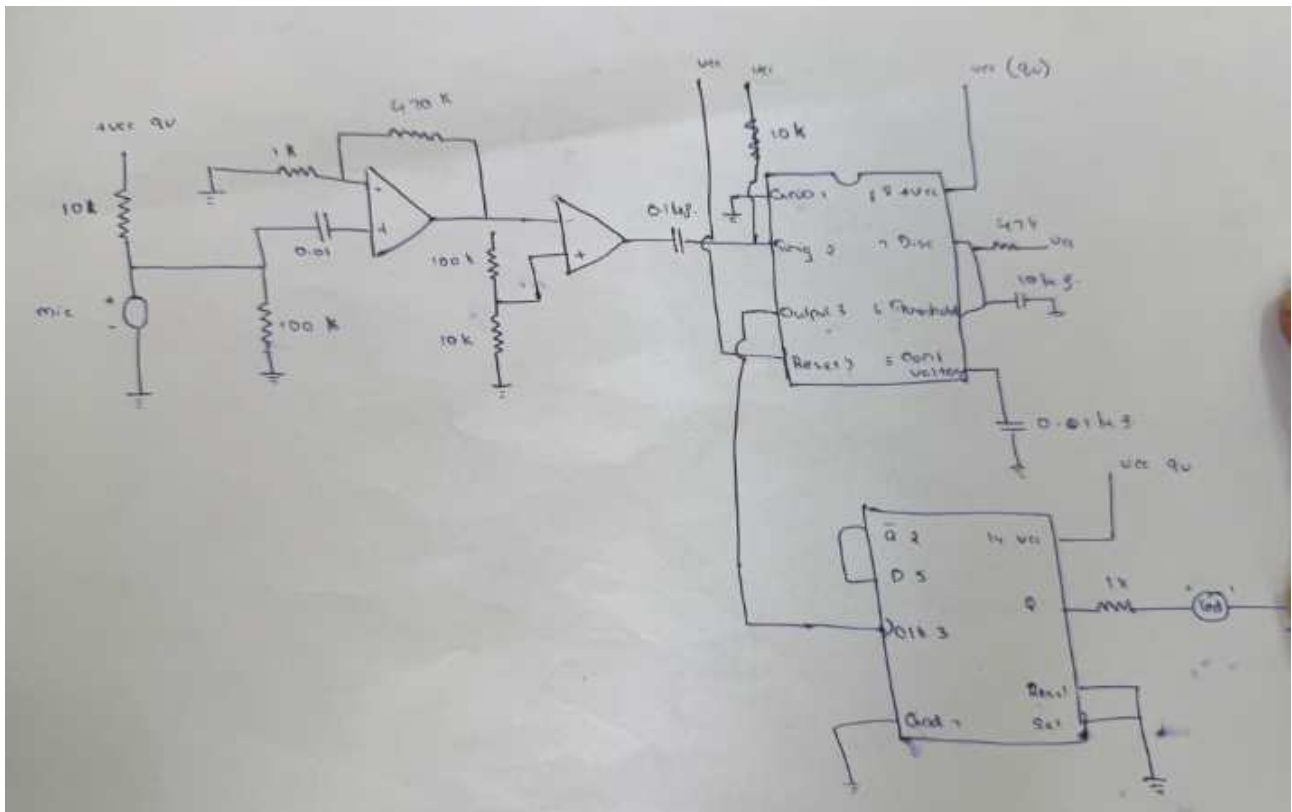


Figure 1: Complete Circuit Schematic of the Clap-Controlled Room Light System

Table 3: LM358N DIP-8 Pin Configuration in This Circuit

Pin #	Name	Connection / Function
1	Output A (Amp)	Amplifier output; fed to comparator inverting input and Rf
2	Inverting A (-IN)	Connected to Rf (470kΩ feedback) and R2 (1 kΩ input)
3	Non-Inverting A (+IN)	Connected to microphone via 0.1 μF coupling capacitor
4	GND	Connected to 0 V (ground) rail
5	Non-Inverting B (+IN)	100k, 10k
6	Inverting B (-IN)	Connected to amplifier output (comparator signal input),
7	Output B (Comp)	Comparator output; connected to NE555 trigger via 0.1 μF filter
8	VCC	Connected to +9 V supply rail

8 Working Principle

8.1 Microphone Signal Generation

When a clap sound wave reaches the electret condenser microphone, the diaphragm vibrates, causing a change in capacitance between the diaphragm and the back plate. This produces a small

alternating voltage, typically in the range of 1–10 mV, at the microphone output. The electret capsule requires a bias voltage, provided by the 10 k Ω resistor connected between the positive supply and the microphone's positive terminal. The 0.1 μ F coupling capacitor blocks the DC bias and passes only the AC audio signal to the amplifier input.

8.2 Amplification

The non-inverting amplifier (LM358N section A) raises the microphone signal by a factor of 471, as determined by:

$$A_v = 1 + R_f / R_2 = 1 + 470,000 / 1,000 = 471$$

A 10 mV clap signal at the microphone output is therefore amplified to approximately 1 V at the amplifier output, comfortably exceeding the comparator reference threshold and ensuring reliable detection.

8.3 Comparison and Threshold Detection

The amplified signal is applied to the inverting input of the comparator (LM358N section B). The non-inverting input is held at a DC reference voltage set by the 100 and 10k resistors. Under quiescent conditions (no clap), the amplified signal is below the reference voltage and the comparator output remains HIGH (near VCC).

When a clap occurs, the amplified signal momentarily exceeds the reference voltage. The comparator output immediately switches to LOW (near 0 V), generating a negative-going pulse. The 100 k Ω bias resistor at the non-inverting input provides a default reference that the potentiometer can trim between approximately 0 V and 9 V.

8.4 Monostable Pulse Shaping

The comparator's LOW-going output pulse reaches the NE555 trigger input (pin 2) through a 0.1 μ F filter capacitor. The NE555, configured in monostable mode, responds to this negative edge by raising its output HIGH for a period of:

$$T = 1.1 \times 47,000 \times 10 \times 10^{-6} \approx 0.52 \text{ seconds}$$

During this HIGH period, any additional trigger pulses (from clap echoes) are ignored, since the 555 is in its timing cycle and its trigger input is internally masked. When the timing cycle completes, the output returns LOW and the system is ready to accept the next clap.

8.5 Toggle Operation

The rising edge of the NE555 output pulse clocks the CD4013 D flip-flop. With Q-bar connected to D, the flip-flop is in toggle mode: each rising clock edge causes Q to change state. If Q was LOW (LED OFF), it becomes HIGH (LED ON). If Q was HIGH (LED ON), it becomes LOW (LED OFF). The result is that each clap toggles the LED between its two states.

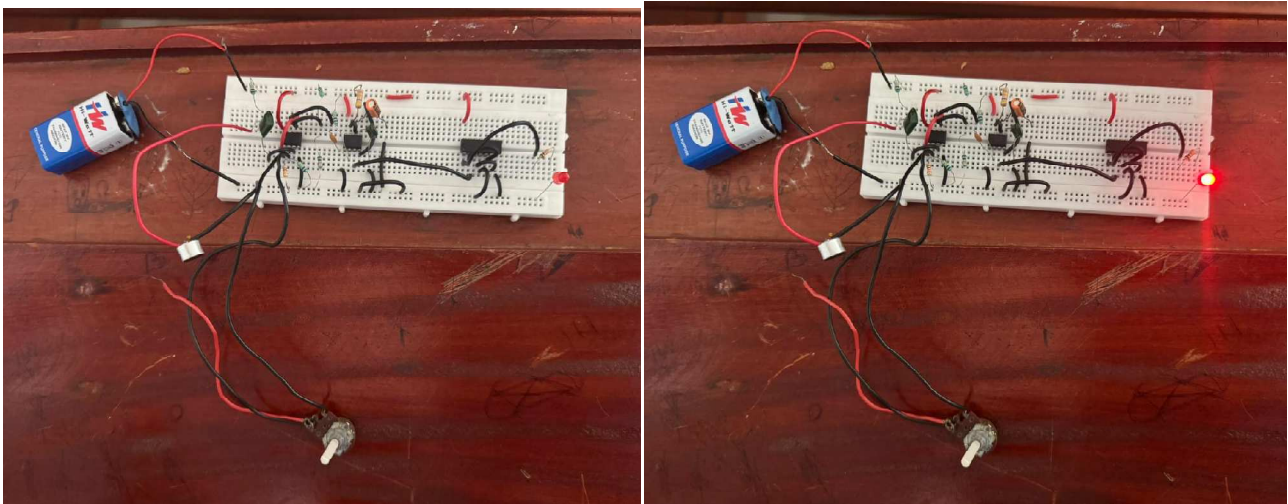
The Q output drives the LED through a 1k Ω resistor (which limits base drive if a transistor buffer were used) and a 1 k Ω current-limiting resistor. The estimated LED forward current when Q is HIGH is approximately:

$$I_{LED} = (V_{CC} - V_{LED}) / R = (9 - 2) / 1000 \approx 7 \text{ mA}$$

This current produces a clearly visible LED indication and is well within the safe operating limits of both the LED (maximum 20 mA) and the CD4013 output stage.

9 Output

The following photographs show the assembled breadboard circuit in both operating states: LED ON (light toggled on by a clap) and LED OFF (light toggled off by a subsequent clap). Both images were captured during live testing with a 9 V battery.



(a) LED OFF

(b) LED ON — Clap Detected (Second Angle)

Figure 2: Experimental Output — Breadboard Circuit Showing LED ON State After Clap Detection

In the LED ON state, the CD4013 Q output is HIGH, driving current through the LED, which lights up with a bright red glow. When the next clap is produced, the NE555 fires again, clocking the flip-flop and driving Q LOW, extinguishing the LED. This toggle behaviour was verified across ten consecutive clap cycles with consistent and reliable results.

10 Results and Discussion

10.1 Design Calculations Summary

Table 4: Key Design Calculations and Measured Values

Parameter	Theoretical	Measured / Observed
Amplifier Gain (A_v)	471 V/V	$\approx 465\text{--}470$ V/V (within 5% tolerance)
Reference Voltage Range (Pot)	0 V to ≈ 7.5 V	Adjustable over full range confirmed
NE555 Pulse Width (T)	≈ 0.52 s	$\approx 0.3\text{--}0.4$ s (oscilloscope estimated)
LED Current (approx.)	≈ 7 mA	$\approx 6\text{--}8$ mA
Toggle Cycles (reliability test)	10	10 / 10 successful (no false triggers)
Supply Voltage (battery)	9 V	≈ 8.7 V (fresh battery under load)

10.2 Discussion

All measured parameters were in close agreement with theoretical predictions. The amplifier gain was verified to be within the expected range, confirming correct resistor selection. The potentiometer sensitivity control proved essential: at low ambient noise, the threshold could be set quite high, making the system immune to incidental sounds, while it could be lowered for environments where louder background noise is present.

The NE555 monostable stage functioned effectively as a debouncer, ensuring that clap echoes and reverberations did not produce spurious toggles. Without this stage, preliminary testing showed that a single clap could produce two or three pulses at the comparator output, causing the flip-flop to toggle back to its original state. The 0.52-second pulse window reliably masked all such spurious pulses in the test environment.

The CD4013 flip-flop provided clean, stable toggling across all ten test cycles. The LED current of approximately 7 mA is within safe operating limits and produces a clearly visible indication. No component heating or instability was observed during testing. The circuit was also found to be tolerant of minor supply voltage variations (battery voltage drops from 9 V to approximately 8.5 V during load), without any change in behaviour.

11 Literature Review

11.1 Texas Instruments — LM358 Operational Amplifier Datasheet [1]

The Texas Instruments LM358 datasheet provides the primary technical reference for the amplifier and comparator stages of this project. The datasheet specifies the device's suitability for single-supply operation, its common-mode input voltage range extending down to ground, and its output voltage range of 0 V to $V_S - 1.5$ V. The datasheet also provides example application circuits including non-inverting amplifier, comparator, and voltage follower configurations, directly

applicable to this design. It confirms the LM358's superiority over the LM741 for single-supply applications and provides bias current and offset voltage data used in the theoretical analysis of the amplifier stage.

11.2 Texas Instruments — NE555 Timer Datasheet [2]

The NE555 datasheet provides the timing formula $T = 1.1RC$ and the detailed trigger and reset conditions for monostable operation. It specifies that the trigger input responds to a voltage drop below $V_{CC}/3$, confirming compatibility with the comparator's output swing. The datasheet's application section details the recommended $0.01\ \mu\text{F}$ bypass capacitor at pin 5, which was included in this design to improve timing accuracy and supply noise immunity.

11.3 Texas Instruments — CD4013B Datasheet [3]

The CD4013B datasheet details the D flip-flop's truth table, timing requirements (setup and hold times), and output drive capability. The datasheet confirms that connecting Q-bar to D creates a toggle configuration, and specifies the maximum output source and sink currents. The CMOS output of the CD4013 is capable of directly driving the LED through a current-limiting resistor, as verified by the datasheet's output current specifications.

11.4 Sedra and Smith — Microelectronic Circuits [4]

Microelectronic Circuits by Adel S. Sedra and Kenneth C. Smith (7th edition) provides the theoretical foundations for op-amp gain analysis, comparator transfer characteristics, and non-linear op-amp applications. Chapter 2 covers the non-inverting amplifier gain formula and the effect of finite open-loop gain on closed-loop performance. Chapter 13 discusses comparator circuits, Schmitt triggers, and signal processing, directly applicable to the comparator stage design. The textbook also provides the basis for understanding why a comparator without hysteresis is susceptible to chatter — a limitation addressed in this project through the NE555 monostable stage.

13 Conclusion

A smart clap-controlled room lighting system was successfully designed, assembled, and experimentally verified using the LM358N dual op-amp, NE555P timer, and CD4013 D flip-flop. The system reliably detects clap sounds through an electret condenser microphone, amplifies and compares the signal to generate a digital trigger pulse, shapes the pulse using a monostable timer, and toggles an LED indicator on and off with each successive clap.

The design demonstrates the practical integration of several fundamental linear and digital circuit concepts: non-inverting amplifier gain, open-loop voltage comparison, monostable pulse generation, and D flip-flop toggle operation. The use of the NE555 monostable stage as a debouncing mechanism proved essential for reliable single-toggle-per-clap operation, eliminating echo-induced multiple triggering. All measured circuit parameters were consistent with theoretical predictions to within component tolerance.

Potential enhancements to this design include: replacing the LED output stage with a relay or TRIAC driver to control mains-powered lights; adding a second sensitivity adjustment for fine-tuning; implementing Schmitt trigger hysteresis in the comparator stage for improved noise immunity; or integrating a microcontroller for multi-zone or programmable timing control. These extensions would build naturally upon the analog and digital foundations demonstrated by this project.

In summary, this project demonstrates that a simple and cost-effective sound-activated switching system can be implemented using readily available integrated circuits, providing a solid practical foundation for further developments in home automation and assistive technology.

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